AI PRODUCT MANAGEMENT 2 · DETAILED COURSE

Prompt Engineering and Context Engineering

Writing and managing the instructions and context that make model behaviour reliable in production.

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CHAPTER 13

Prompt Engineering and Context Engineering

Writing and managing the instructions and context that make model behaviour reliable in production.

Prompt Engineering Fundamentals

Giving clear task, context, constraints and format so the model produces the output you need.

A good prompt states the task, context, constraints and output format. Clear instructions reduce variation and make outputs easier to evaluate.

Prompt Structure

Typical parts: role, task, context, examples, constraints and output format.

Typical parts are role, task, context, examples, constraints and output format. Keep the order consistent so teams can read and edit prompts quickly.

System Prompts

Persistent instructions that set role, rules and behaviour for a whole conversation.

A system prompt sets persistent role, rules and behaviour for a whole conversation or product. It is the first place to put product policy, tone and safety rules.

User Prompts

The per-turn request or input from the user.

The user prompt is the per-turn request. Treat its content as untrusted input, especially when it is combined with company data.

Role-Based Prompting

Assigning a persona or expertise to shape tone and focus.

Asking the model to act as an expert, such as a senior tax advisor, can shape tone and focus. It does not add knowledge the model lacks, so verify the content.

Zero-Shot Prompting

Asking for the task with no examples.

Zero-shot asks for the task with no examples. It is the fastest approach and works well for common tasks with clear instructions.

One-Shot Prompting

Providing a single example of the desired output.

One-shot gives a single example of the desired output. It helps the model match format or style.

Few-Shot Prompting

Providing several examples so the model infers the pattern.

Few-shot provides several examples so the model infers the pattern. Choose examples that cover typical and difficult cases.

Instruction-Based Prompting

Explicit step-by-step instructions and rules for the task.

Explicit step-by-step instructions and rules define the task. Numbered steps and clear do and do-not lists reduce ambiguity.

Chain-of-Thought Concepts

Asking the model to reason through steps, which can improve accuracy on multi-step problems.

Asking the model to reason through steps can improve accuracy on multi-step problems. For products, consider whether the reasoning should be shown to users or kept internal.

Prompt Chaining

Splitting a task into several prompts where each output feeds the next.

A task is split into several prompts where each output feeds the next, such as extract, then validate, then summarise. Chains are easier to test than one very long prompt.

Structured Prompting

Using sections, delimiters and schemas to make prompts unambiguous.

Sections, delimiters and schemas make prompts unambiguous. Use clear markers around user-provided text so the model does not confuse it with instructions.

JSON-Based Outputs

Requesting JSON with defined fields and validating it in code.

Requesting JSON with defined fields lets software use the result. Validate it in code and handle malformed output with a retry or a fallback.

Prompt Templates

Reusable prompts with placeholders filled at run time.

Templates are reusable prompts with placeholders filled at run time. They keep wording consistent across features and make changes easier to review.

Dynamic Prompt Variables

Inserting user, document or session data into templates safely.

Variables insert user, document or session data into a template. Escape or delimit inserted text, so it cannot change the instructions.

Context Engineering

Deciding exactly what information, tools and memory the model sees at each step.

Context engineering decides exactly what information, tools and memory the model sees at each step. It often matters more than the wording of the prompt.

Context Window Optimisation

Prioritising, summarising and trimming content to fit limits while keeping what matters.

Prioritise, summarise and trim content to fit the window while keeping what matters. Put the most important information where the model attends most reliably, usually near the instructions or the question.

Memory and Context Management

Storing and recalling conversation facts, summaries and user preferences across turns and sessions.

Memory stores conversation facts, summaries and preferences across turns and sessions. Ask users what should be remembered and let them delete it.

Prompt Versioning

Tracking prompt changes like code, with versions, owners and test results.

Track prompt changes like code, with versions, owners and test results. Without versioning, teams cannot tell which change caused a quality drop.

Prompt Evaluation

Scoring prompt versions against a fixed test set before release.

Score prompt versions against a fixed test set before release. Use the same inputs for every version so the comparison is fair.

Prompt Injection Awareness

Untrusted text can try to override instructions; treat external content as data, not commands.

Untrusted text, including content from web pages or documents, can try to override instructions. Treat external content as data, not commands, and limit what tools the model can use.

Prompt Guardrails

Input and output checks, refusals and policy rules that keep behaviour within bounds.

Input and output checks, refusals and policy rules keep behaviour within bounds. Guardrails are layers, so combine rules, classifiers and human review for high-risk cases.

Production Prompt Management

Central storage, review, rollout, monitoring and rollback of prompts used in live products.

Production prompts need central storage, review, staged rollout, monitoring and rollback. Treat a prompt change like a code change with the same release discipline.

Summary of terms

Term	Meaning
Prompt Engineering Fundamentals	Giving clear task, context, constraints and format so the model produces the output you need.
Prompt Structure	Typical parts: role, task, context, examples, constraints and output format.
System Prompts	Persistent instructions that set role, rules and behaviour for a whole conversation.
User Prompts	The per-turn request or input from the user.
Role-Based Prompting	Assigning a persona or expertise to shape tone and focus.
Zero-Shot Prompting	Asking for the task with no examples.
One-Shot Prompting	Providing a single example of the desired output.
Few-Shot Prompting	Providing several examples so the model infers the pattern.
Instruction-Based Prompting	Explicit step-by-step instructions and rules for the task.
Chain-of-Thought Concepts	Asking the model to reason through steps, which can improve accuracy on multi-step problems.
Prompt Chaining	Splitting a task into several prompts where each output feeds the next.
Structured Prompting	Using sections, delimiters and schemas to make prompts unambiguous.
JSON-Based Outputs	Requesting JSON with defined fields and validating it in code.
Prompt Templates	Reusable prompts with placeholders filled at run time.
Dynamic Prompt Variables	Inserting user, document or session data into templates safely.
Context Engineering	Deciding exactly what information, tools and memory the model sees at each step.
Context Window Optimisation	Prioritising, summarising and trimming content to fit limits while keeping what matters.
Memory and Context Management	Storing and recalling conversation facts, summaries and user preferences across turns and sessions.
Prompt Versioning	Tracking prompt changes like code, with versions, owners and test results.
Prompt Evaluation	Scoring prompt versions against a fixed test set before release.
Prompt Injection Awareness	Untrusted text can try to override instructions; treat external content as data, not commands.
Prompt Guardrails	Input and output checks, refusals and policy rules that keep behaviour within bounds.
Production Prompt Management	Central storage, review, rollout, monitoring and rollback of prompts used in live products.

Practice exercise: Write a system prompt with a role, rules and JSON output, then version it and test five variations on a fixed set of inputs.

Self-check

1. Explain Prompt Engineering Fundamentals in two sentences to a non-technical stakeholder, then say what you would measure for it.
2. Explain Structured Prompting in two sentences to a non-technical stakeholder, then say what you would measure for it.
3. Explain Production Prompt Management in two sentences to a non-technical stakeholder, then say what you would measure for it.

Related SkillNavora material

- Prompt and Context Engineering: 13 classes with templates.
- AI Skills Academy: Every AI skills course in one place.
- AI Tutor: Ask questions on any topic.
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